الآي بي والتشفير في زمن التزوير

Lecture 3 — Layer 3: IPv4 and IPsec Security and Protection

> **Layer 3 (Network layer)** : moves packets **between networks**. IPv4 delivers packets; ACLs, filtering and IPsec add security.

> **IPv4** : a **connectionless** network-layer protocol with **32-bit** addresses written in dotted decimal (192.168.10.25).

Routers forward mainly by the **destination** address. The **source** address is only a claim — not strong proof of identity.

Q/ What are the IPv4 header fields and their role ?

Field	Role
Version	Always 4 for IPv4
IHL	Header length (where data begins)
DSCP/ECN	Priority / quality of service
Total Length	Header + data size
Identification, Flags, Fragment Offset	Fragmentation and reassembly
TTL	Decreases by 1 at each router; at 0 the packet is discarded (prevents endless loops)
Protocol	Upper layer: ICMP, TCP, UDP, **ESP**, **AH**
Header Checksum	Detects errors in the header
Source / Destination	Claimed sender (32 bits) / intended receiver (32 bits)

Q/ Why does IPv4 need extra security ?

- It was designed for **delivery** only: it doesn't encrypt contents or strongly authenticate the source address.

IPv4 threats

Threat	What can happen
IP spoofing	A false source address is placed in a packet
Packet interception	Unencrypted data read on an untrusted path
ICMP abuse	Ping sweeps, reconnaissance, floods, false control messages
DoS / DDoS	Bandwidth, router or server resources exhausted
Fragmentation attacks	Odd fragments confuse filters or consume reassembly resources
Routing attacks	Traffic redirected, dropped or intercepted

> **IP spoofing** : faking the source IPv4 address to **hide the sender, imitate another device, bypass IP-based restrictions, or support DoS**.

> **ICMP** : diagnostics and error reporting — ping uses **Echo Request / Echo Reply**; traceroute also depends on ICMP.

Topic	Explanation	Protection
Ping sweep	ICMP to many addresses to find live hosts	Permit only required ICMP, rate-limit, log
DoS / DDoS	Floods make a service unavailable	Filtering, rate limiting, monitoring, upstream protection
Fragmentation	Tiny/overlapping/incomplete fragments confuse devices	Drop malformed fragments, reassemble/inspect, update

Don't blindly block all ICMP — some messages are needed for diagnostics and correct operation.

Q/ How are routers protected against routing attacks ?

- Strong admin authentication, **SSH**, restricted management addresses, **routing-protocol authentication**, updates, backups, logging, disabling unnecessary services.

Protection mechanisms

> **ACL (Access Control List)** : an **ordered** set of permit/deny rules checking source, destination, protocol, port or ICMP type.

Standard ACL	Extended ACL
Mainly checks the source IPv4 address	Checks **source, destination, protocol and TCP/UDP port**
e.g. permit 192.168.10.0/24	e.g. permit HTTPS from 192.168.20.0/24 to 172.16.10.10

Q/ What are the ACL processing principles ?

1. Rules are checked **in order**.
2. The **first matching rule** is applied.
3. **Specific rules before general** rules.
4. Permit only required traffic; test before deployment.

> **Ingress filtering** : checks traffic **entering** the network; rejects impossible or suspicious source addresses.

> **Egress filtering** : checks traffic **leaving**; stops internal systems from using unauthorized source addresses.

> **uRPF (Unicast Reverse Path Forwarding)** : accepts a packet only if its source is reachable through the **expected return path**.

> **Network segmentation** : separates students, staff, administration, servers and management networks.

No single Layer 3 control stops every attack — combine them.

IPsec

> **IPsec** : protects IP communication **at the network layer**, useful across untrusted networks like the Internet. Adds encryption, integrity, authentication and **anti-replay**.

Component	Main role
AH (Authentication Header)	Integrity, data-origin authentication, anti-replay — **no confidentiality**
ESP (Encapsulating Security Payload)	**Encryption** (confidentiality) + integrity, authentication, anti-replay
SA (Security Association)	How traffic is protected: algorithms, keys, peer, lifetime, **SPI**, anti-replay info
IKEv2	Authenticates peers, negotiates parameters, establishes keys, creates/manages SAs

Feature	AH	ESP
Confidentiality	**No**	**Yes**
Integrity	Yes	Yes (when configured)
Source authentication	Yes	Yes (when configured)
Anti-replay	Yes	Yes
VPN use	Limited	**Commonly used**

Feature	Transport mode	Tunnel mode
Protects	Mainly the original **payload**	The **entire original packet**
IP header	Original header stays visible	A **new outer IP header** is added
Typical use	Host-to-host	**Gateway-to-gateway, site-to-site VPN**, remote access
Overhead	Lower	Higher

Two university buildings linked over the Internet → **tunnel mode** (with ESP).

> **Security Association (SA)** : a **one-directional** logical agreement — two-way traffic needs **one SA in each direction**.

Q/ What does IKEv2 do ?

1. Authenticates the two peers.
2. Negotiates security algorithms and parameters.
3. Establishes keys and creates IPsec SAs.
4. Maintains the relationship and re-keys when needed.

Q/ Site-to-site example (192.168.10.0/24 ↔ 192.168.20.0/24) — what do the gateways do ?

- Identify protected traffic → authenticate each other → establish SAs → encrypt packets → decrypt at the other site.

Requirement	Recommended protection
Students must not access router management	Management ACL + segmentation
Internet packet claims an internal source	Ingress filtering + uRPF
Only required traffic to the grade server	Extended ACL
Two buildings exchange private data over the Internet	IPsec tunnel mode
Suspicious ICMP traffic	Rate limiting + logging + monitoring
Routing information must be protected	Routing authentication + router hardening

Key points: IPv4 has no built-in confidentiality or strong source authentication; **ESP** is the choice when confidentiality is required (AH doesn't encrypt); **tunnel mode** protects the whole packet; **IKEv2** builds and manages SAs.

Summary of Network Security L3: Layer 3: IPv4 & IPsec Security. Built only from the lecture file; check the original slides for anything you're unsure about.